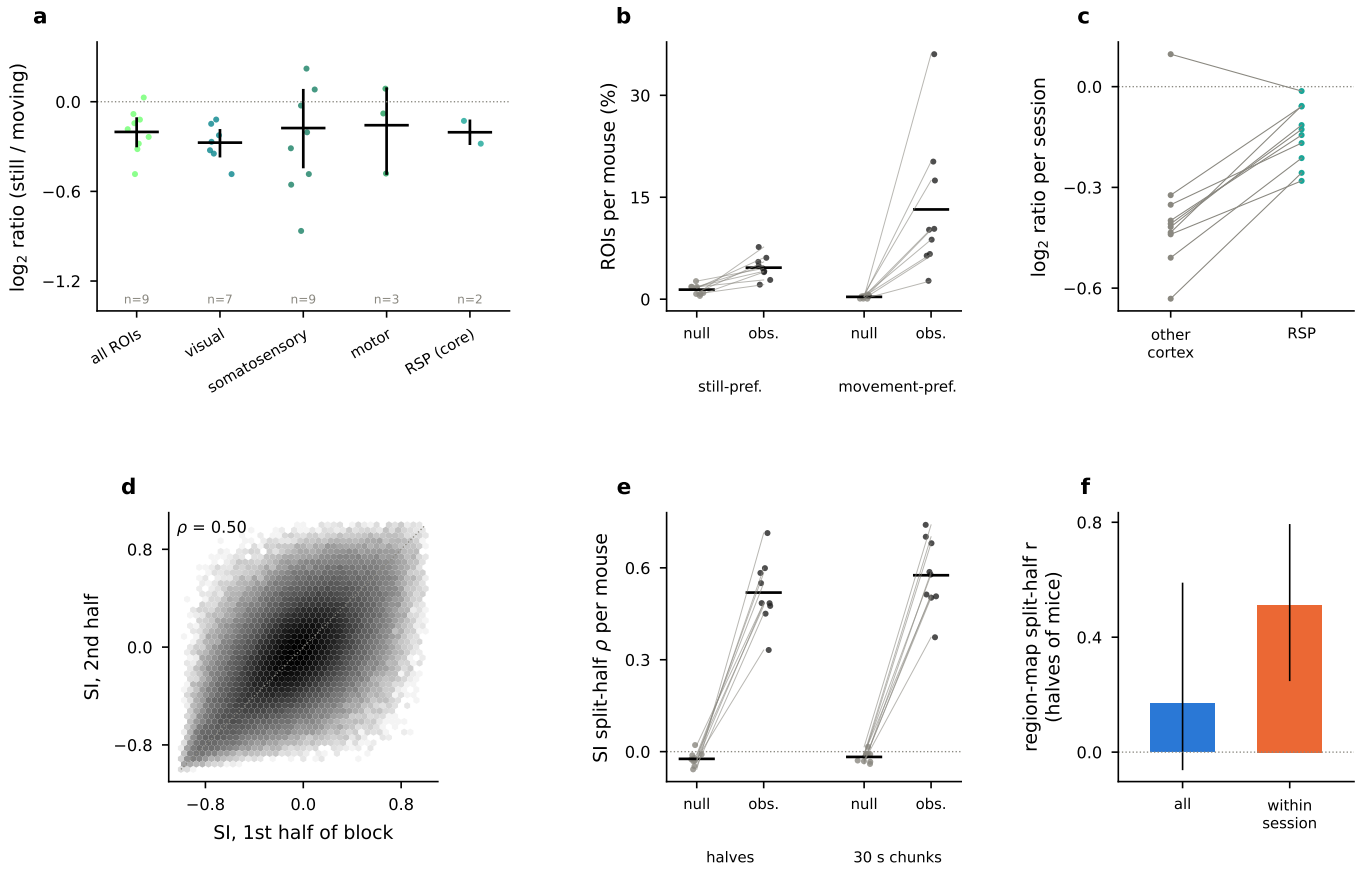


Figure 17: Correlates of stillness in the IBL mesoscope: lower activity in stillness across dorsal cortex, and a stillness preference that is reproducible within recordings. **a** Activity in still against moving 1 s bins of the passive block before the task replay (grey screen and receptive-field mapping; median 6.2 min per session), $\log_2((\text{still}+e)/(\text{moving}+e))$ of each ROI's deconvolved activity with e = 2% of its mean; ROIs averaged per session (sessions with $\geq$20 ROIs in the group), sessions per mouse. Moving: lick or face motion energy above the session's 60th percentile (the wheel is not recorded in this block); still: no movement within $\pm$1 s and $\leq$30 s after the last movement. Groups by Beryl region (Allen colours); black, mean and 95% bootstrap CI over mice; n, mice. **b** Still- and movement-preferring ROIs per mouse (Welch test of still against moving bins, at most one of each per 2 s, $p < 0.01$), observed against a circular-shift null (activity rotated against the behavioural labels, 100 offsets $\geq$60 s). Bars: means over mice. One session whose null exceeded 5% (slow drift; a quarter of its ROIs silent in all still bins) is excluded from all panels. **c** RSP against the other imaged ROIs of the same session (lines: sessions). **d** Reproducibility within recordings: stillness index SI = (still $-$ moving)/(still $+$ moving) of each ROI in the first against the second half of the passive block (each half $\geq$10 still and $\geq$10 moving bins; 335,204 ROIs, 46 sessions; log colour scale). **e** Spearman ρ of SI between the two halves, and between alternating 30 s chunks, per session averaged per mouse (black), against a null in which the second part's activity is rotated against its state labels (20 offsets $\geq$15 s; grey). Lines: mice. **f** Reliability of region maps of the $\log_2$ ratio over animals: Spearman r between region means from two random halves of the mice (200 splits, median and 5–95% range), from raw ROI values (blue) and after subtracting each session's mean (orange); 16 Beryl regions with $\geq$20 ROIs from $\geq$3 mice. Region within session explains 4.2% of ROI variance (label shuffle 0.1%), session 4.1%.